FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 07

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Recap of previous lecture

CONIC SECTION



Topics to be covered

"AP/GP/AGP"



ARITHMETIC PROGRESSION

Progression – those sequences in which there exist certain pattern known as Prog.

for eg. $1, 8, 27, 64, 125, \ldots$

$= 1^3, 2^3, 3^3, 4^3, 5^3, 6^3, 7^3, \ldots$ Progressions.

AP – if $[a_1, a_2, a_3, a_4 \ldots$ are in AP then $[a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \ldots]]$

& this difference is called Common Difference denoted by 'd'

Def^n: – let 1st term $= a$ & C.Diff $= d$ then AP can be represented as;

$a, \underset{1^{st}}{a}, \underset{2^{nd}}{a+d}, \underset{3^{rd}}{a+2d}, \underset{4^{th}}{a+3d}, \underset{5^{th}}{a+4d}, \ldots, \underset{n^{th}\text{ term}}{a+(n-1)d}, \ldots = l$

eg: 1, 3, 5, 7, 9, 11, 13, 15 ⇒ YES it is AP ⟨ a = 1, d = 2

eg 5, 9, 13, 17, 21, 25, 29 ⇒ YES, it is also AP ⟨ a = 5, d = 4

eg 4, 7, 10, 13, 17, 20, 23 - - - ? No it is not AP.

d ≠ 3

① General term / n^{th} term ⇒ $\boxed{t_n = a+(n-1)d} = d.n + (a-d)$,

② Last term ⇒ if last term is l, then we can write, $\boxed{l = a+(n-1)d}$

Note ⇒ ① if given sequence is an AP ⟺ n^{th} term must be linear func^n of n i.e $t_n = An+B$

& in that case C.Diff $\boxed{d = \text{coeff of } 'n'}$

for eg if $t_n = 3n-2$, Can we say that given Seq. is an AP?

(YES, if is the nth of an A.P.) & C.Diff 'd' = 3 & $t_{10} = 30-2 = 28$

(2) In an AP, Sum of terms equidistant from the beginning and end is always constant and is equal to Sum of 1st & last term i.e

$$a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = a_4 + a_{n-3} = \text{- - - - - -}$$

{where Series is $a_1, a_2, a_3, a_4 \text{- - - -} a_{n-2}, a_{n-1}, a_n$}

(3) n^{th} term from end $\leftarrow$ (if Total terms are m) $\rightarrow$

$= [m-(n-1)]^{th}$ term from beginning

$= (m-n+1)^{th}$ term from beginning $= t_{m-n+1} = \boxed{a + (m-n)d}$

Eg: 7th term from last if T-Term are 15 = ?

(M-I) 1 2 3 4 5 6 7 8 (9) 10 11 12 13 14 15

9 — 7th term from Last

10 11 12 13 14 15 — 6 terms

(M-II) $m = 15$, $n = 7$ so $[m-(n-1)] = 15-(7-1) = 15-6 = 9$

i.e. 7th term from Last = 9th term from beginning

or n^{th} " " " = $(m-n+1)^{th}$ " " "

(*) $\boxed{n^{th} \text{ term from last} = a+(m-n)d}$

If last term is given then n^{th} term from end = $\boxed{l+(n-1)(-d)}$

Sum of n terms :- $S_n = a_1 + a_2 + a_3 + a_4 + \cdots + a_{n-2} + a_{n-1} + a_n$

again $S_n = a_n + a_{n-1} + a_{n-2} + \cdots + a_3 + a_2 + a_1$

$$2S_n = (a_1 + a_n) + (a_2 + a_{n-1}) + (a_3 + a_{n-2}) + \cdots + (a_{n-1} + a_2) + (a_n + a_1)$$

$$= \underbrace{(a_1 + a_n) + (a_1 + a_n) + (a_1 + a_n) + \cdots + (a_1 + a_n) + (a_1 + a_n)}_{n \text{ pair}}$$

$$2S_n = n(a_1 + a_n) \Rightarrow S_n = \frac{n}{2}(a_1 + a_n)$$

i.e. $\boxed{S_n = \frac{n}{2}[a + l] = \frac{n}{2}[2a + (n-1)d]}$ where $l = a + (n-1)d$

Note :- In an AP, Sum of n terms must be of the type, $S_n = An^2 + Bn$ & $\boxed{C.Diff = 2(\text{Coeff of } n^2)}$

Q: which term of the sequence

4, 9, 14, 19 - - - - is 124

Sol: $a = 4,\ d = 5,\ t_n = 124$

$t_n = a + (n-1)d$

$124 = 4 + (n-1)5$

$120 = 5n - 5 \Rightarrow 5n = 125$

$n = 25^{th}$

Q: Determine the number of terms in the seq, 3, 7, 11. - - -, 407

also find 20th term.

Q: How many terms are there in the sequence

3, 6, 9, 12 - - - - . 111 ?

$a = 3,\ d = 3,\ l = 111$

$l = a + (n-1)d$

$111 = 3 + (n-1)3$

$111 = 3n \Rightarrow n = \frac{111}{3} = 37$

$l = a + (n-1)d$

$407 = 3 + (n-1)4$

$408 = 4n$

$n = 102$

$t_{20} = a + (20-1)d$

$= 3 + 19(4)$

$= 3 + 76$

$= 79$

Q. Find the sum of series upto 20 terms

$1+4+7+10+\text{----}$ 20 terms

$a=1,\ d=3,\ n=20,$

$S_n=\frac{n}{2}[2a+(n-1)d]$

$S_{20}=\frac{20}{2}[2\times1+(20-1)3]$

$=10[2+57]=590$

Q. Evaluate

$5+13+21+\text{----}+181$

$a=5,\ d=8,\ l=181$

W.K.T that $l=a+(n-1)d$

$181=5+(n-1)8 \Rightarrow n=\frac{184}{8}=23$

$S_{23}=\frac{n}{2}[a+l]$

$=\frac{23}{2}[5+181]$

$=23\times93$

Arithmetic Mean → if a, b, c are in AP then b is called AM of a & c

w.k.that $(b-a)=(c-b) \Rightarrow \boxed{b=\frac{a+c}{2}}$

⊛ three numbers are in AP if $\boxed{2b=a+c}$

Insertion of n AM'S b/n a & b → Let $a, \underbrace{A_1, A_2, A_3, \ldots, A_n}_{\text{AM'S of } a \& b}, b$ are in AP

In this AP, T. Terms $=(n+2)$

& $b=a+(n+2-1)d$

$\boxed{d=\frac{b-a}{n+1}}$

& $A_1=a+d$

$A_2=a+2d$

$A_3=a+3d \ldots\ldots$

Q Insert three AM's b/n 3 & 19

Sol: $3, A_1, A_2, A_3, 19$ are in AP

$n=5$, $a=3$, $l=19$

$l = a+(n-1)d$

$19 = 3+(5-1)d \Rightarrow d=4$

$A_1 = a+d = 3+4 = 7$

$A_2 = a+2d = 3+8 = 11$

$A_3 = a+3d = 3+12 = 15$

i.e Req. Seq is $3, 7, 11, 15, 19$ Ans.

Selection of terms in an AP →

3 Terms: $a-d, a, a+d$, C.Diff $= d$

5 terms: $a-2d, a-d, a, a+d, a+2d$, C.Diff $= d$

4 terms: $a-3d, a-d, a+d, a+3d$, C.Diff $= 2d$

Eg: the Sum of 3 Nos in an AP is -3 & Product is 8 then Nos are?

Sol: Let $(a-d)+a+(a+d) = -3 \Rightarrow 3a = -3 \Rightarrow a = -1$

again $(a-d)(a)(a+d) = 8 \Rightarrow (1-d^2)(-1) = 8 \Rightarrow d = \pm 3$

$d=3$ then Nos $= -4, -1, 2$, $d=-3$, Nos $= 2, -1, -4$

Geometric Progression (G.P)

eg 2, 6, 18, 54 ---- it is GP ∵ Common Ratio (r) $= \frac{6}{2} = \frac{18}{6} = \frac{54}{18} = \cdots$

Def^n $a, ar, ar^2, ar^3, \ldots, ar^{n-1}$ where 1st term $= a$ & Common Ratio $= r$

1st, 2nd, 3rd, 4th, ..., n^{th} term

i.e $r = \frac{t_2}{t_1} = \frac{t_3}{t_2} = \frac{t_4}{t_3} = \cdots$ is in General, $r = \frac{t_n}{t_{n-1}}$

(*) n^{th} term: $t_n = ar^{n-1}$

(*) Last term: $l = ar^{n-1}$

(*) n^{th} term from end = (if Total term $= m$)

(I) $(m-n+1)^{th}$ term from beginning $= t_{m-n+1} = a \cdot r^{m-n}$

(II) n^{th} term from end $= l\left(\frac{1}{r}\right)^{n-1}$

Sum of n terms: i.e $S_n = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1}$

$-\quad rS_n = \quad ar + ar^2 + ar^3 + \cdots + ar^{n-1} + ar^n$

$(1-r)S_n = a - ar^n \Rightarrow \boxed{S_n = \frac{a(1-r^n)}{1-r}}$ or $\boxed{S_n = \frac{a(r^n-1)}{r-1}}$

Note: ① $S_n = \frac{a(1-r^n)}{1-r} = \frac{a-ar^n}{1-r} = \frac{a-(a\cdot r^{n-1})\cdot r}{1-r} = \boxed{\frac{a-lr}{1-r}}$ or $\boxed{\frac{lr-a}{r-1}}$

② Consider 2, 4, 8, 16, 32, 64, 128, 256 i.e In any GP "Product of the terms equidistant from the beginning and the end is always constant and is equal to Product of 1st & last term, i.e where $l = ar^{n-1}$

$\boxed{t_1 \cdot t_n = t_2 t_{n-1} = t_3 t_{n-2} = \cdots\cdots}$

Q: 9th term of the series

$\frac{1}{4}, \frac{-1}{2}, 1, -2, \ldots$?

$a = \frac{1}{4}$, $r = \frac{t_2}{t_1} = \frac{-1/2}{1/4} = -2$

$n = 9$ so $t_n = ar^{n-1}$

$\Rightarrow t_9 = ar^{9-1}$

$= \frac{1}{4}(-2)^8$

$= \frac{1}{4} \times 256 = 64$

Q: which term of the GP $5, 10, 20, 40, \ldots$ is 5120 ?

Sol: $a = 5$, $r = \frac{t_2}{t_1} = \frac{10}{5} = 2$, $l = 5120$

$l = ar^{n-1}$

$5120 = 5(2)^{n-1}$

$1024 = 2^{n-1}$

$2^{10} = 2^{n-1}$

$n - 1 = 10 \Rightarrow n = 11$

i.e. $t_{11} = 5120$

Q: the 4th term from end of the GP 3, 6, 12, 24, ---3072

Sol: $r = \frac{6}{3} = 2$

So n^{th} term from end $= l\left(\frac{1}{r}\right)^{n-1}$

$= 3072\left(\frac{1}{2}\right)^{4-1}$

$= 384$

Q: the sum of 7 terms of 3, 6, 18----

$n=7, a=3, r=\frac{6}{3}=2$

$S_7 = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}$

$= \frac{3(2^7-1)}{2-1} = 3(128-1) = 3\times127$

Q: Evaluate $2+6+18+\cdots+4374$

$S = \frac{lr-a}{r-1} = \frac{4374\times3-2}{3-1}$

Q: Evaluate $5+55+555+5555+\ldots\ldots+n$ terms

Sol: $S_n = 5+55+555+5555+\ldots\ldots t_n$

$S_n = 5[1+11+111+1111+\ldots\ldots n \text{ terms}]$

$= \frac{5}{9}[9+99+999+9999+\ldots\ldots n \text{ terms}]$

$= \frac{5}{9}[(10-1)+(10^2-1)+(10^3-1)+\ldots\ldots n \text{ terms}]$

$= \frac{5}{9}[(10+10^2+10^3+\ldots n \text{ terms}) - (\underbrace{1+1+1+\ldots}_{n \text{ terms}})]$

$a = 10$, $r = 10$

$= \frac{5}{9}\left[\frac{a(r^n-1)}{r-1} - n\right]$

$= \frac{5}{9}\left[\frac{10(10^n-1)}{10-1} - n\right]$

$= \frac{5}{9}\left[\frac{10}{9}(10^n-1) - n\right]$ //

Sum of an Infinite GP → $S_\infty = a + ar + ar^2 + ar^3 + \ldots\ldots\infty$ terms; $-1 < r < 1$

Sol: $S_n = \frac{a(1-r^n)}{1-r} = \frac{a - ar^n}{1-r} = \frac{a}{1-r} - \frac{ar^n}{1-r}$

when $n \to \infty$ then $\boxed{S_\infty = \frac{a}{1-r}} - 0$ $\left(\because\ r^n \to 0 \text{ when } n \to \infty\right)$

ie Conclusion:

$$\boxed{S_\infty = \begin{cases} \frac{a}{1-r}, & |r| < 1 \\ \infty, & \text{otw} \end{cases}}$$

Sol: $S = 6^{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots\ldots\infty}$

$a = \frac{1}{2}$; $r = \frac{1/4}{1/2} = \frac{1}{2}$

$S = 6^{\frac{a}{1-r}} = 6^{\frac{1/2}{1-1/2}} = 6^{\frac{1}{1}} = 6^1 = 6$

Q Evaluate $6^{\frac{1}{2}} \cdot 6^{\frac{1}{4}} \cdot 6^{\frac{1}{8}} \ldots\ldots\infty$ terms ?

PQ: Find the sum upto ∞ term of GP; $\frac{-5}{4}, \frac{5}{16}, \frac{-5}{64}$ ..... ∞ term?

Ans: -1

$$S_\infty = \frac{a}{1-r} = \text{---} = -1$$

Q: The sum of ∞ GP is 8 and 2nd term is 2 then find 1st term?

Sol: ATQ, $S_\infty = 8$ & $a_2 = 2$

$$\frac{a}{1-r} = 8 \quad \& \quad ar = 2$$

$$r = \frac{2}{a}$$

$$\frac{a}{1-\frac{2}{a}} = 8$$

$$\frac{a^2}{a-2} = 8 \Rightarrow a^2 - 8a + 16 = 0 \Rightarrow (a-4)^2 = 0 \Rightarrow a = 4$$

Selection of terms in GP →

No of terms	Terms	C. Ratio
3	$\frac{a}{r}, a, ar$	r
5	$\frac{a}{r^3}, \frac{a}{r}, a, ar, ar^3$	r
4	$\frac{a}{r^2}, \frac{a}{r}, ar, ar^2$	r^2
6	$\frac{a}{r^4}, \frac{a}{r^2}, \frac{a}{r}, ar, ar^2, ar^4$	r^2

Q. If the sum of 3 terms in GP is 38 and Product is 1728 then Nos are ?

Sol. Let the Nos are $\frac{a}{r}, a, ar$

ATQ, $P = 1728$

$\left(\frac{a}{r}\right)(a)(ar) = 1728$

$a^3 = 1728 \Rightarrow a = 12$

Again; $\frac{a}{r} + a + ar = 38$

$\Rightarrow r = \frac{3}{2}$ or $\frac{2}{3}$

So Nos are $\Rightarrow$ 8, 12, 18

18, 12, 8

Geometric Mean:- if a, b, c are in GP then b is called GM of a & c and it is given as $\boxed{GM = \pm\sqrt{ac}}$ $\left(\because \frac{b}{a} = \frac{c}{b} \Rightarrow b^2 = ac\right)$

Ex) three numbers a, b, c are in GP if $b^2 = ac$

→ if a, c are +ve we will take +ve sign.
& if a & c are −ve " " " −ve sign } Imp.

e.g. GM of 4 & 9 = ? $= \pm\sqrt{4\times 9} = +6$

e.g. GM of −9 & −4 = ? $= -\sqrt{(-4)(-9)} = -\sqrt{36} = -6$

e.g. GM of −9 & 4 = ? = N.D

Q: if AM & GM of two numbers are 34 & 16 resp then find nos.

Sol: Let α & β are the Numbers then ATQ.

$AM = 34$ $\qquad$ $GM = 16$

$\frac{\alpha+\beta}{2} = 34$ $\qquad$ $\sqrt{\alpha\beta} = 16$

$\alpha+\beta = 68$ $\qquad$ $\alpha\beta = 256$

$\alpha = 64, \beta = 4$

Arithmetico Geometric Progression (AGP) → it is denoted as;

$$a, (a+d)r, (a+2d)r^2, (a+3d)r^3, \ldots\ldots\ldots, (a+(n-1)d)\cdot r^{n-1}$$

Eg: find $1-3x+5x^2-7x^3+\ldots\ldots\infty$ terms where $|x|<1$, $\boxed{r=-x}$

Sol $S_\infty = 1-3x+5x^2-7x^3+\ldots\ldots\infty$

$-xS_\infty = \quad -x+3x^2-5x^3+\ldots\ldots\infty$

$(1+x)S_\infty = 1-2x+2x^2-2x^3+2x^4-\ldots\ldots$

$(1+x)S_\infty = 1-2x\left[1-x+x^2-x^3+\ldots\right]$

$= 1-2x\left[\frac{1}{1-(-x)}\right] \quad \left(\because S_\infty = \frac{a}{1-r}\right)$

$(1+x)S = 1-2x\left(\frac{1}{1+x}\right)$

$S_\infty = \frac{1}{1+x} - \frac{2x}{(1+x)^2} = \frac{1-x}{(1+x)^2}$ //

P8: Evaluate $\frac{3}{5}+\frac{7}{5^2}+\frac{11}{5^3}+\frac{15}{5^4}+\cdots\cdots$ ∞ terms.

Sol:

$$r=\frac{1/5^2}{1/5}=\frac{1}{5}$$

$$S=\frac{3}{5}+\frac{7}{5^2}+\frac{11}{5^3}+\frac{15}{5^4}+\cdots\cdots\infty$$

$$-\frac{1}{5}S=\quad\frac{3}{5^2}+\frac{7}{5^3}+\frac{11}{5^4}+\cdots\cdots\infty$$

$$\left(1-\frac{1}{5}\right)S=\frac{3}{5}+\frac{4}{5^2}+\frac{4}{5^3}+\frac{4}{5^4}+\cdots\cdots$$

$$\frac{4}{5}S=\frac{3}{5}+\frac{4}{5^2}\left(1+\frac{1}{5}+\frac{1}{5^2}+\cdots\infty\right)$$

$$\frac{4}{5}S=\frac{3}{5}+\frac{4}{25}\left(\frac{1}{1-\frac{1}{5}}\right)$$

$$=\frac{3}{5}+\frac{4}{25}\left(\frac{5}{4}\right)$$

$$\frac{4}{5}S=\frac{3}{5}+\frac{1}{5}=\frac{4}{5}$$

$$S=1$$


Thank
YOU